

PROJECT 2 STATIC ROUTING

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TASK 1: CREATING STATIC ROUTES IN THE ROUTER

ROUTER 1

R1 IP: 192.50.20.100 – eth0/0

STATIC ROUTE R1:

23.0.0.0/8 12.0.0.2

26.0.0.0/8 12.0.0.2

32.0.0.0/8 12.0.0.2

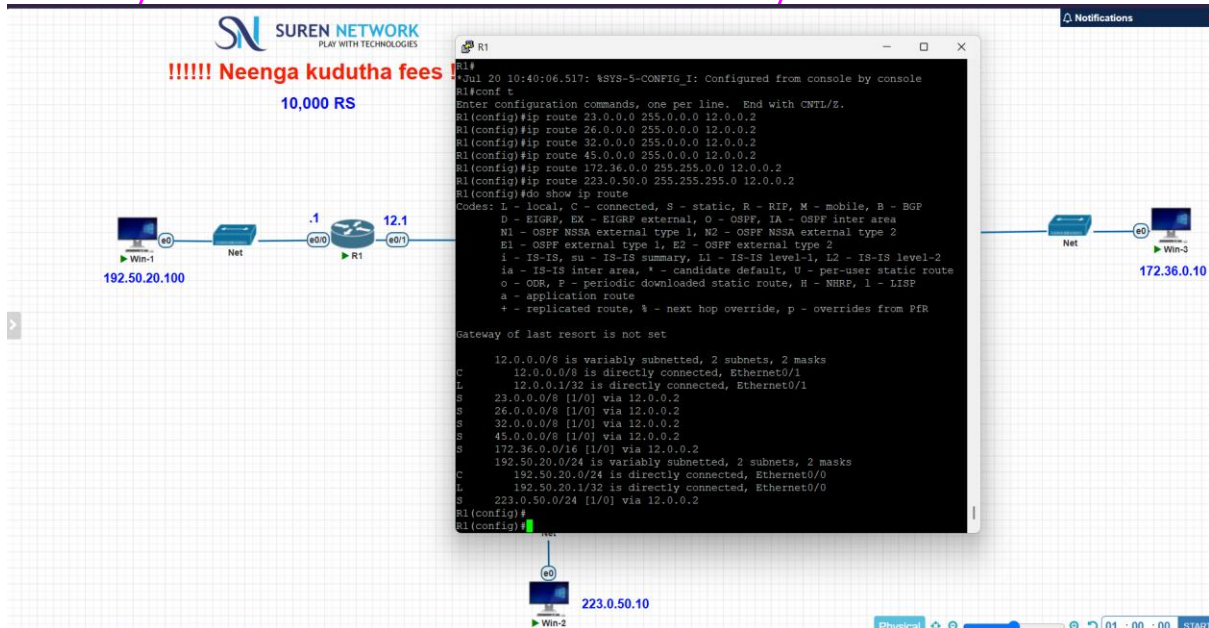
R1 IP: 12.0.0.1 – eth0/1

STATIC ROUTE R1:

45.0.0.0/8 12.0.0.2

172.36.0.0/16 12.0.0.2

223.0.50.0/24 12.0.0.2



ROUTER 2

R2 IP: 12.0.0.1 – eth0/0

R2 IP: 23.0.0.1 – eth0/2

STATIC ROUTE R2:

32.0.0.0/8 23.0.0.2

45.0.0.0/8 24.0.0.2

172.36.0.0/16 24.0.0.2 & 23.0.0.2

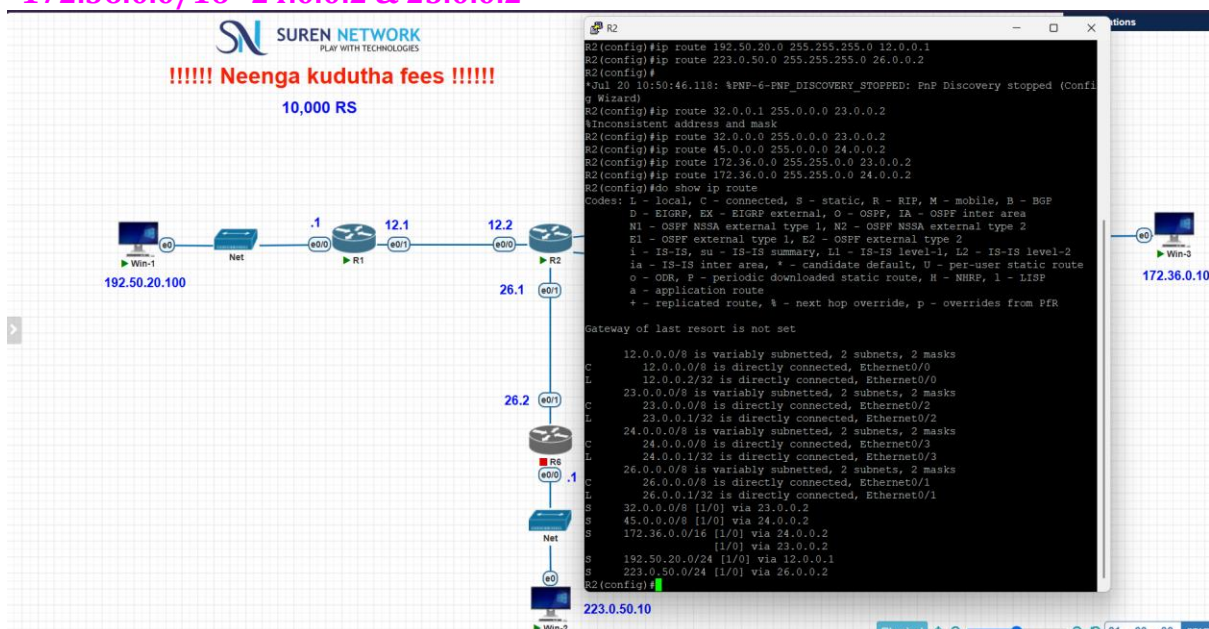
R2 IP: 26.0.0.1 – eth0/1

R2 IP: 24.0.0.1 – eth0/3

STATIC ROUTE R2:

192.50.20.0/24 12.0.0.1

223.0.50.0/24 26.0.0.2



ROUTER 3

R3 IP: 23.0.0.2 – eth0/0

STATIC ROUTE R3:

192.50.20.0/24 23.0.0.1

12.0.0.0/8 23.0.0.1

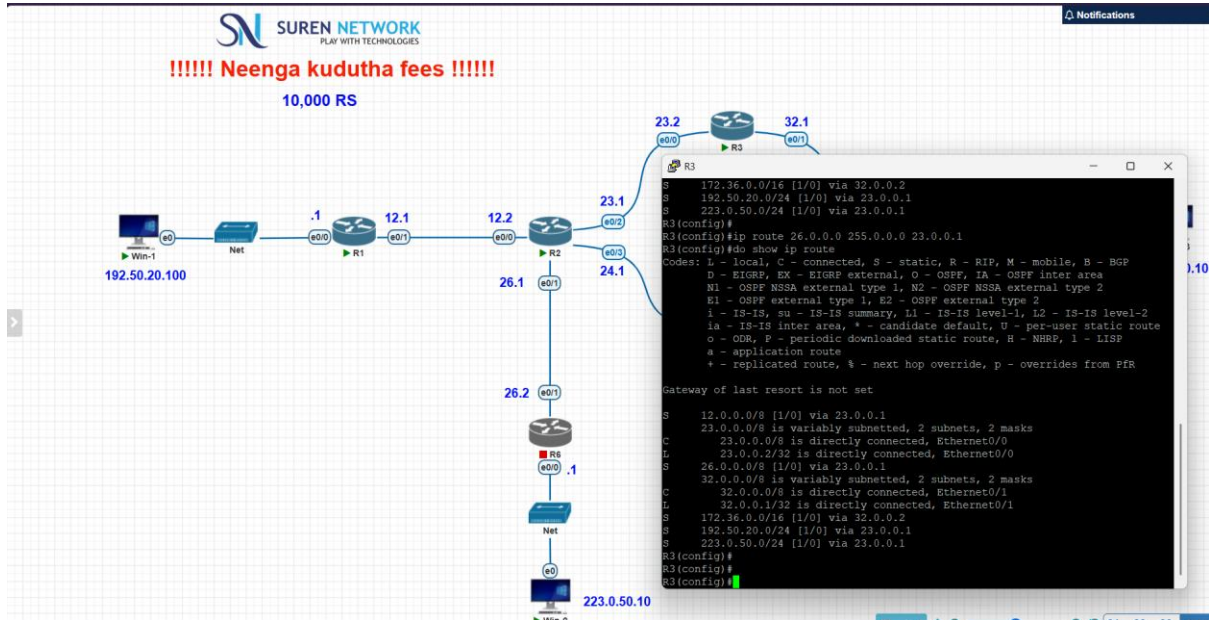
26.0.0.0/8 23.0.0.1

R3 IP: 32.0.0.1 – eth0/2

STATIC ROUTE R3:

172.36.0.0/16 32.0.0.2

223.0.50.0/24 23.0.0.1



ROUTER 4

R4 IP: 24.0.0.2 – eth0/0

STATIC ROUTE R4:

12.0.0.0/8 24.0.0.1

26.0.0.0/8 24.0.0.1

23.0.0.0/8 24.0.0.1

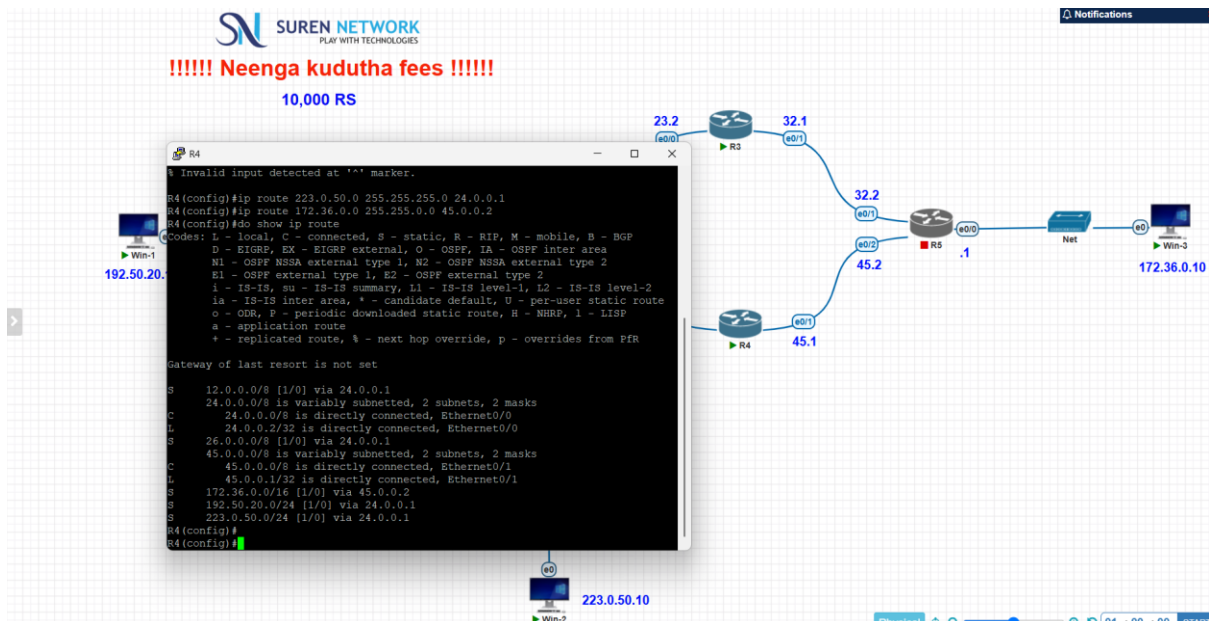
R4 IP: 45.0.0.1 – eth0/1

STATIC ROUTE R4:

172.36.0.0/16 45.0.0.2

192.50.20.0/24 24.0.0.1

223.0.50.0/24 24.0.0.1



ROUTER 5

R5 IP: 172.36.0.1 – eth0/0

R5 IP: 32.0.0.2 – eth0/1

R5 IP: 45.0.0.2 – eth0/2

STATIC ROUTE R5:

12.0.0.0/8 45.0.0.1

23.0.0.0/8 45.0.0.1

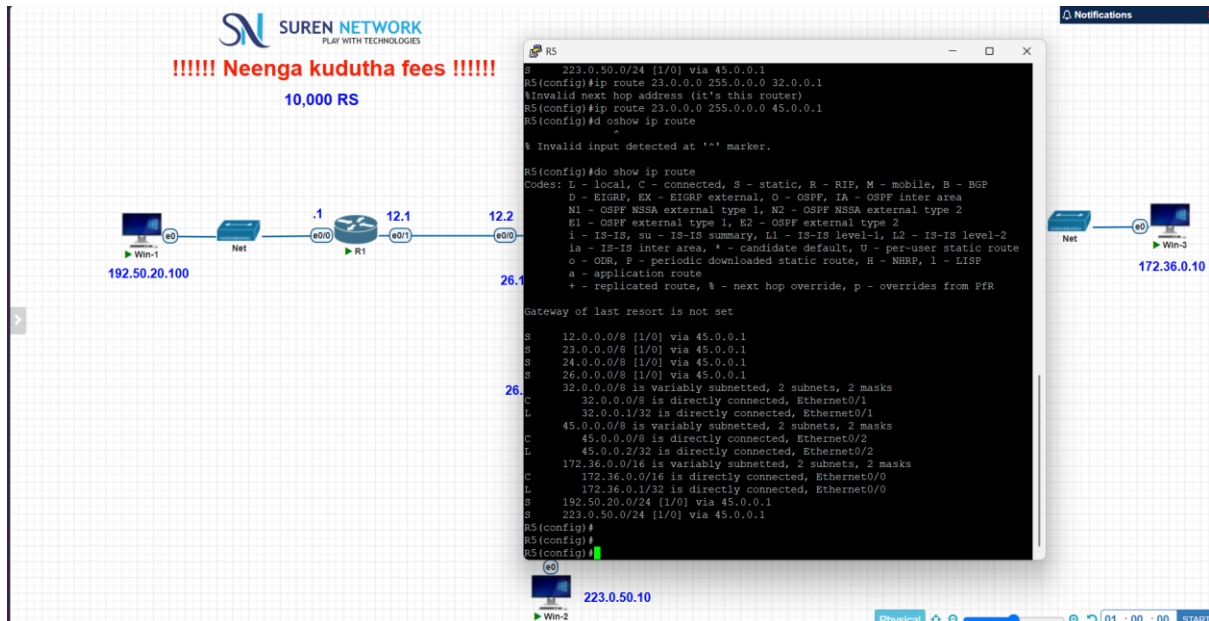
24.0.0.0/8 45.0.0.1

STATIC ROUTE R5:

26.0.0.0/8 45.0.0.1

192.50.20.0/24 45.0.0.1

223.0.50.0/24 45.0.0.1



ROUTER 6

R6 IP: 223.0.50.1 – eth0/0

R6 IP: 26.0.0.2 – eth0/1

STATIC ROUTE R6:

12.0.0.0/8 26.0.0.1

23.0.0.0/8 26.0.0.1

24.0.0.0/8 26.0.0.1

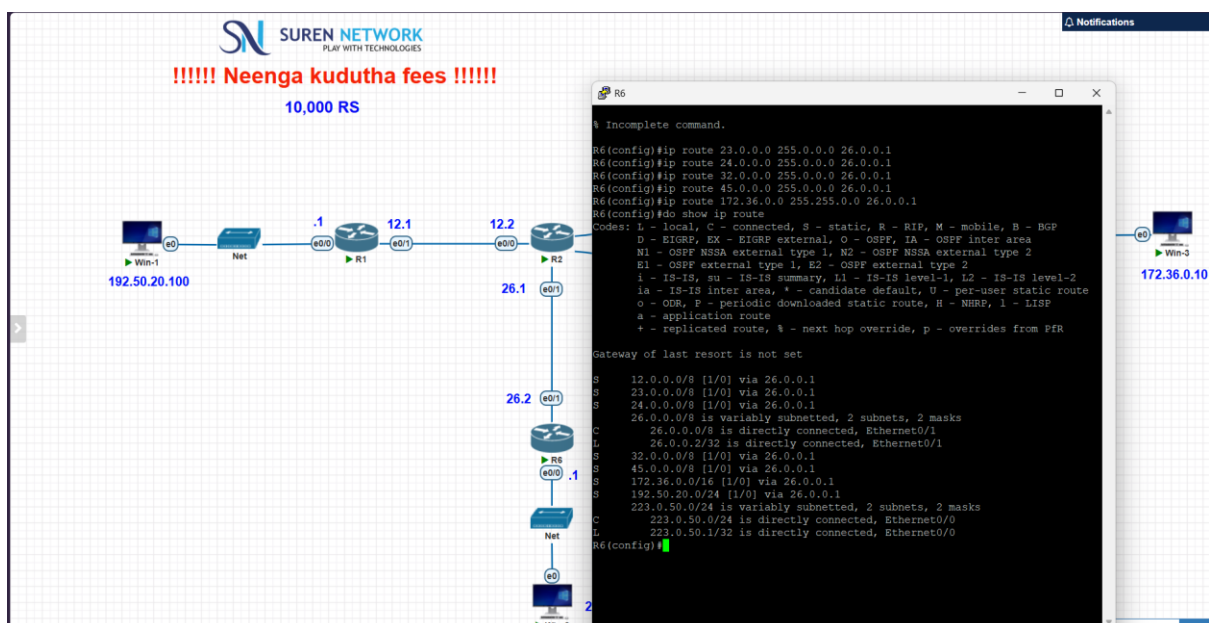
STATIC ROUTE R6:

32.0.0.0/8 26.0.0.1

45.0.0.0/8 26.0.0.1

172.36.0.0/16 26.0.0.1

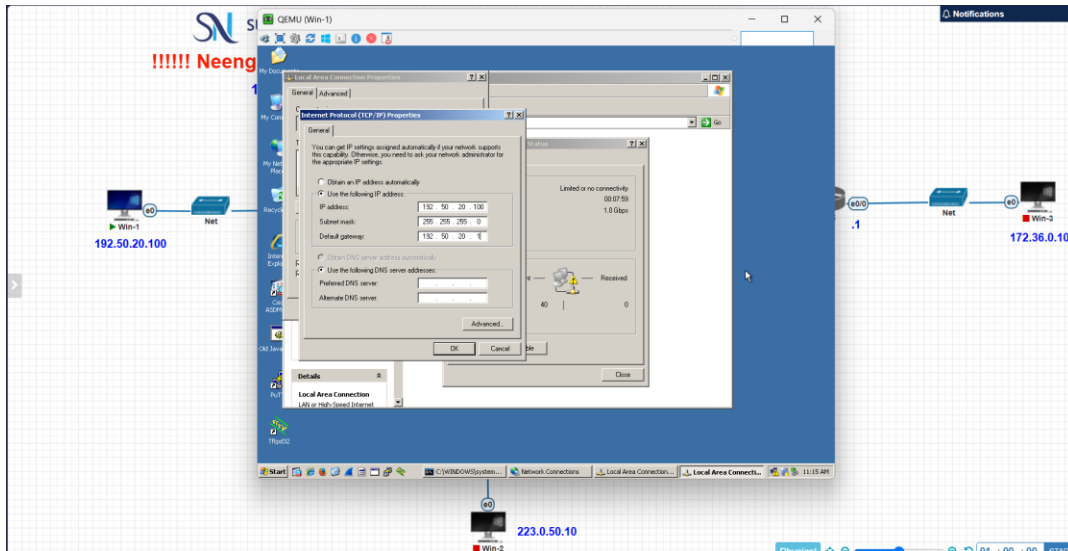
192.50.20.0/24 26.0.0.1



TASK 2: ASSIGNING IP ADDRESS TO PCs

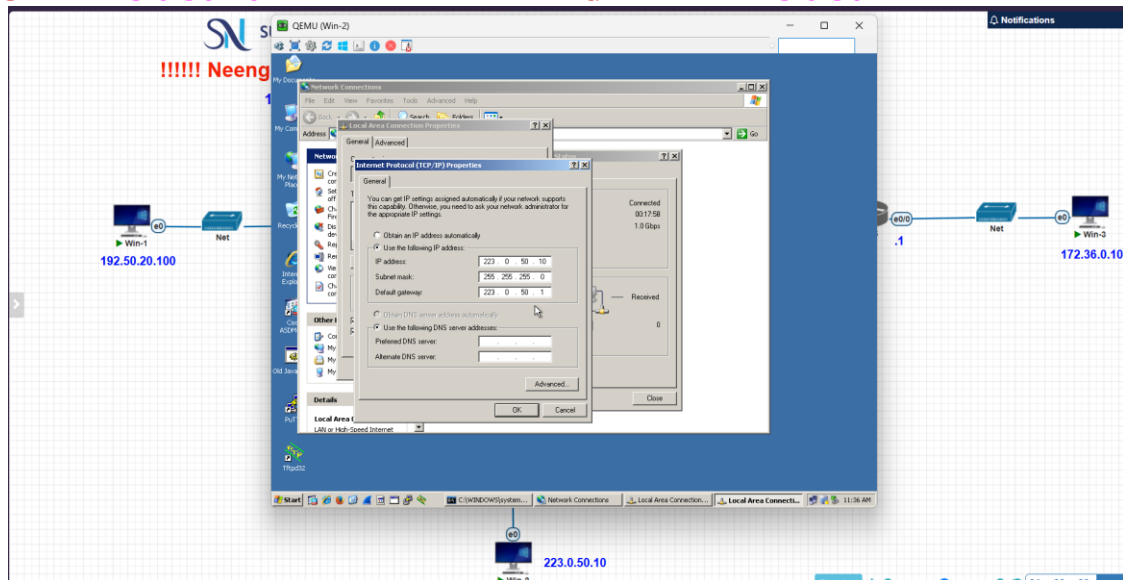
PC1 IP: 192.50.20.100

GATEWAY: 192.50.20.1



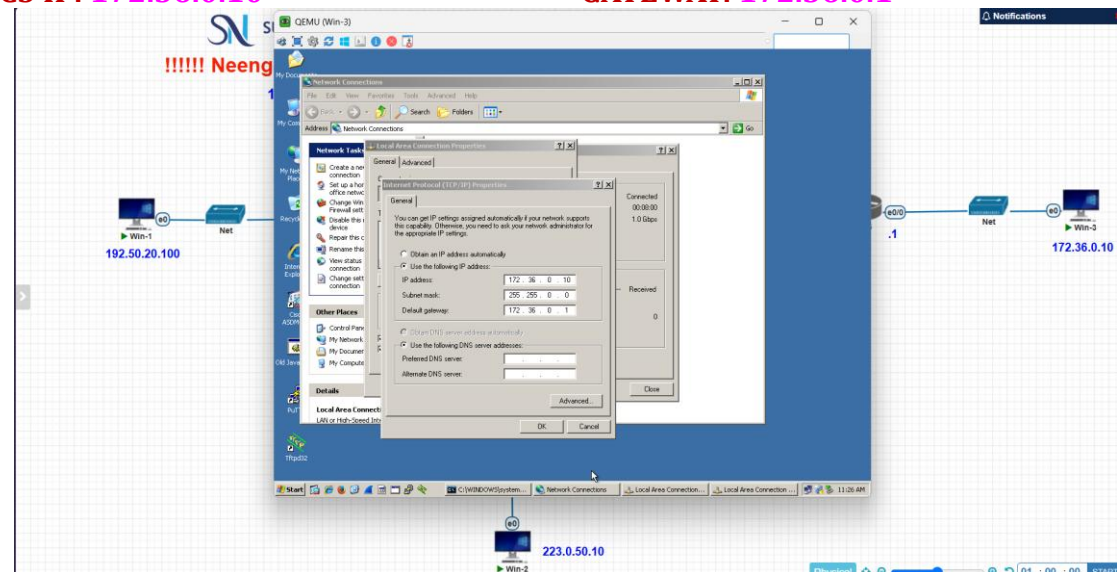
PC2 IP: 223.0.50.10

GATEWAY: 223.0.50.1



PC3 IP: 172.36.0.10

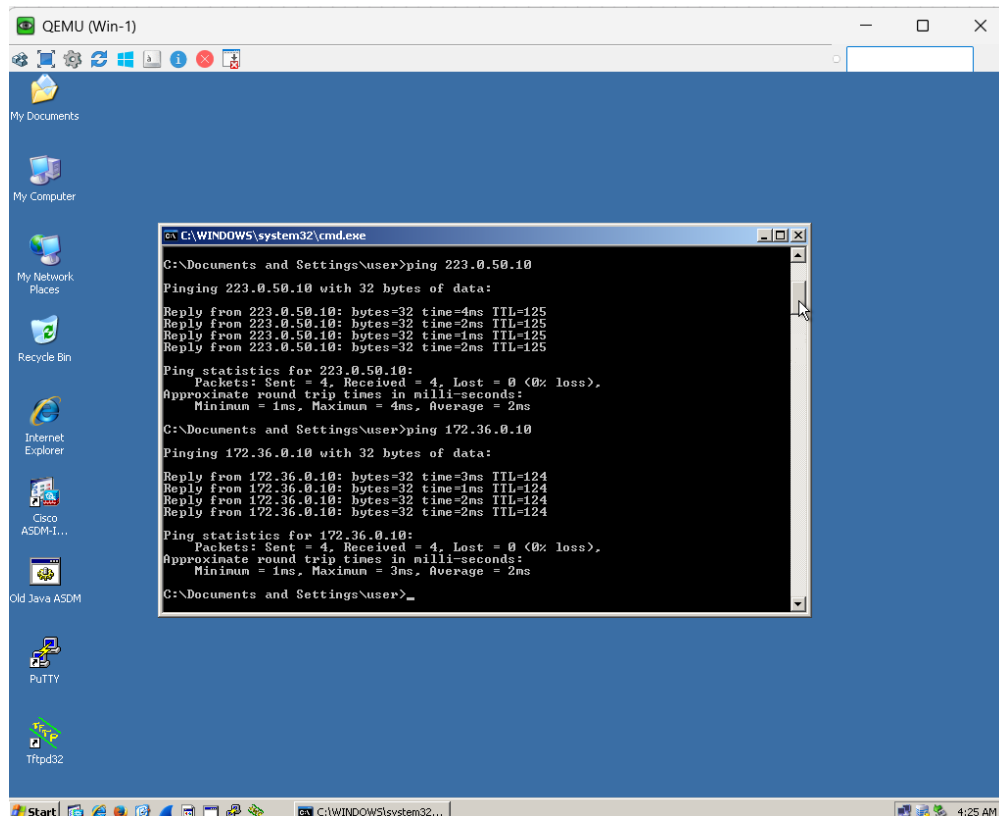
GATEWAY: 172.36.0.1



TASK 3: VERIFYING THE CONNECTION ESTABLISHMENTS

ping from 192.50.20.100 (PC1) to 223.0.50.10 (PC2) &

ping from 192.50.20.100 (PC1) to 172.36.0.10 (PC3)

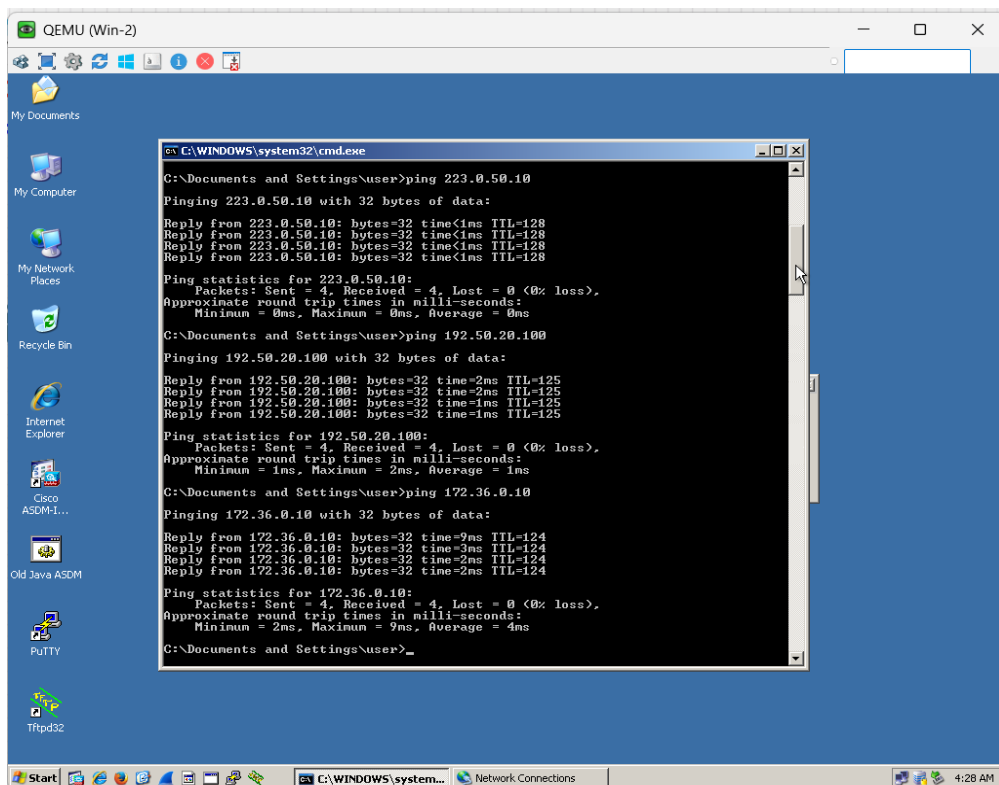


```
C:\WINDOWS\system32\cmd.exe
C:\Documents and Settings\user>ping 223.0.50.10
Pinging 223.0.50.10 with 32 bytes of data:
Reply from 223.0.50.10: bytes=32 time=4ms TTL=125
Reply from 223.0.50.10: bytes=32 time=2ms TTL=125
Reply from 223.0.50.10: bytes=32 time=1ms TTL=125
Reply from 223.0.50.10: bytes=32 time=2ms TTL=125
Ping statistics for 223.0.50.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 4ms, Average = 2ms
C:\Documents and Settings\user>ping 172.36.0.10
Pinging 172.36.0.10 with 32 bytes of data:
Reply from 172.36.0.10: bytes=32 time=3ms TTL=124
Reply from 172.36.0.10: bytes=32 time=1ms TTL=124
Reply from 172.36.0.10: bytes=32 time=2ms TTL=124
Reply from 172.36.0.10: bytes=32 time=2ms TTL=124
Ping statistics for 172.36.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 3ms, Average = 2ms
C:\Documents and Settings\user>
```

ping from 223.0.50.10 (PC2) to 223.0.50.10 (PC2) &

ping from 223.0.50.10 (PC2) to 192.50.20.100 (PC1) &

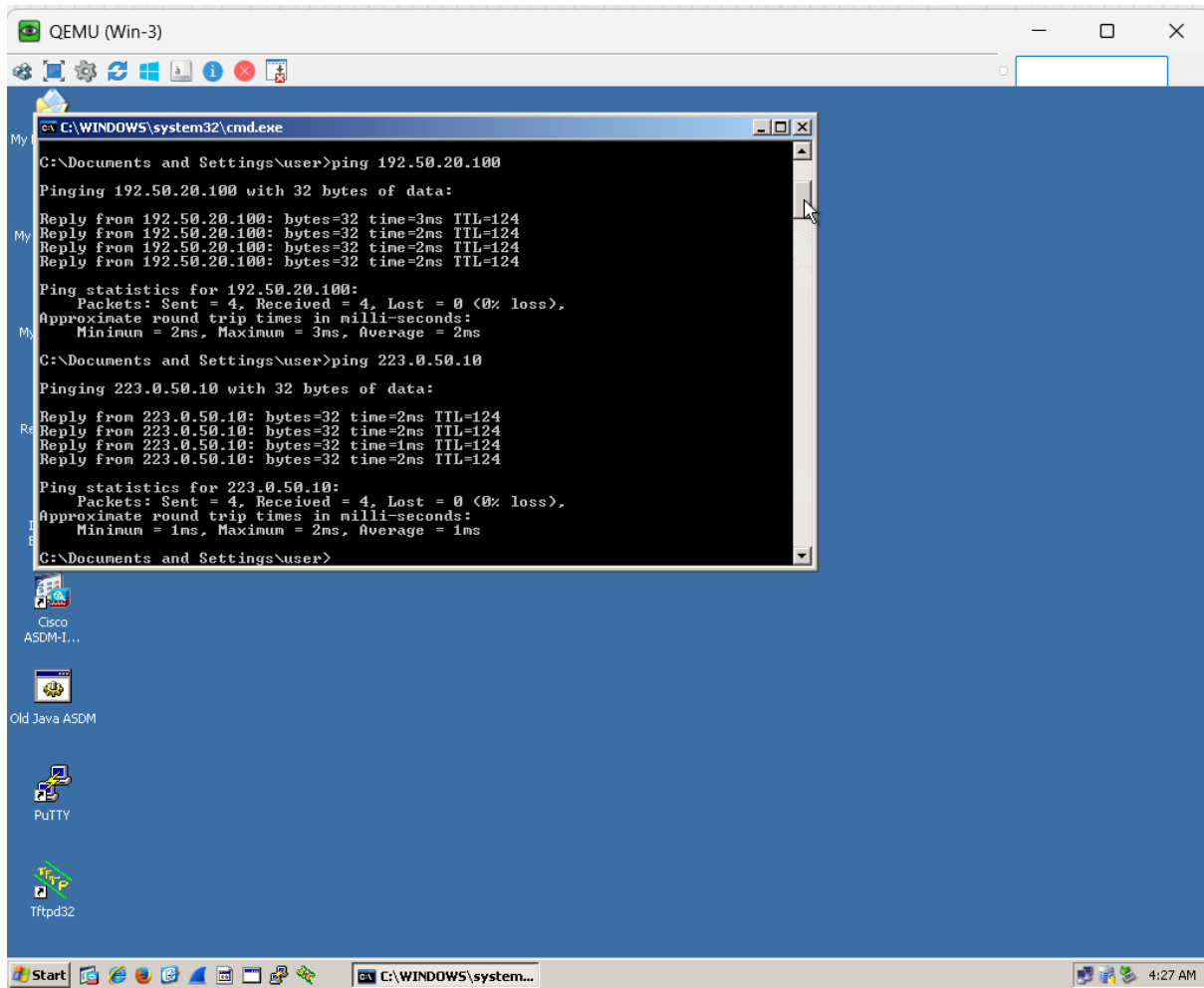
ping from 223.0.50.10 (PC2) to 172.36.0.10 (PC3)



```
C:\WINDOWS\system32\cmd.exe
C:\Documents and Settings\user>ping 223.0.50.10
Pinging 223.0.50.10 with 32 bytes of data:
Reply from 223.0.50.10: bytes=32 time<1ms TTL=128
Reply from 223.0.50.10: bytes=32 time<1ms TTL=128
Reply from 223.0.50.10: bytes=32 time<1ms TTL=128
Reply from 223.0.50.10: bytes=32 time<1ms TTL=128
Ping statistics for 223.0.50.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\Documents and Settings\user>ping 192.50.20.100
Pinging 192.50.20.100 with 32 bytes of data:
Reply from 192.50.20.100: bytes=32 time=2ms TTL=125
Reply from 192.50.20.100: bytes=32 time=2ms TTL=125
Reply from 192.50.20.100: bytes=32 time=1ms TTL=125
Reply from 192.50.20.100: bytes=32 time=1ms TTL=125
Ping statistics for 192.50.20.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms
C:\Documents and Settings\user>ping 172.36.0.10
Pinging 172.36.0.10 with 32 bytes of data:
Reply from 172.36.0.10: bytes=32 time=9ms TTL=124
Reply from 172.36.0.10: bytes=32 time=3ms TTL=124
Reply from 172.36.0.10: bytes=32 time=2ms TTL=124
Reply from 172.36.0.10: bytes=32 time=2ms TTL=124
Ping statistics for 172.36.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 9ms, Average = 4ms
C:\Documents and Settings\user>
```

Ping from 172.36.0.10 (PC3) to 192.50.20.100 (PC1) &

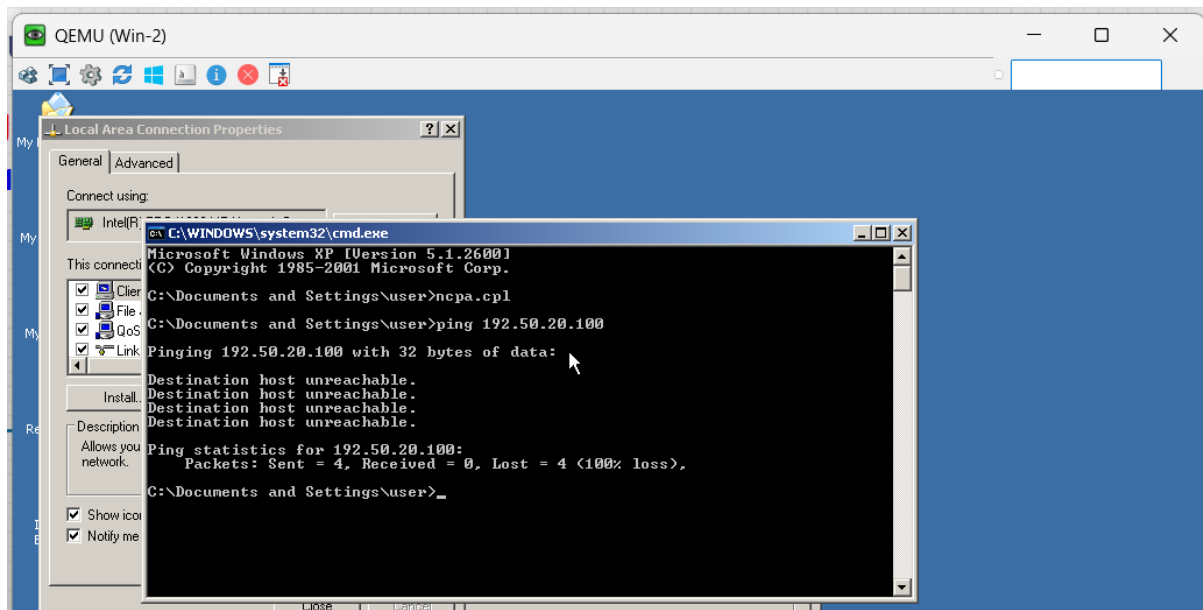
Ping from 172.36.0.10 (PC3) to 223.0.50.10 (PC2)



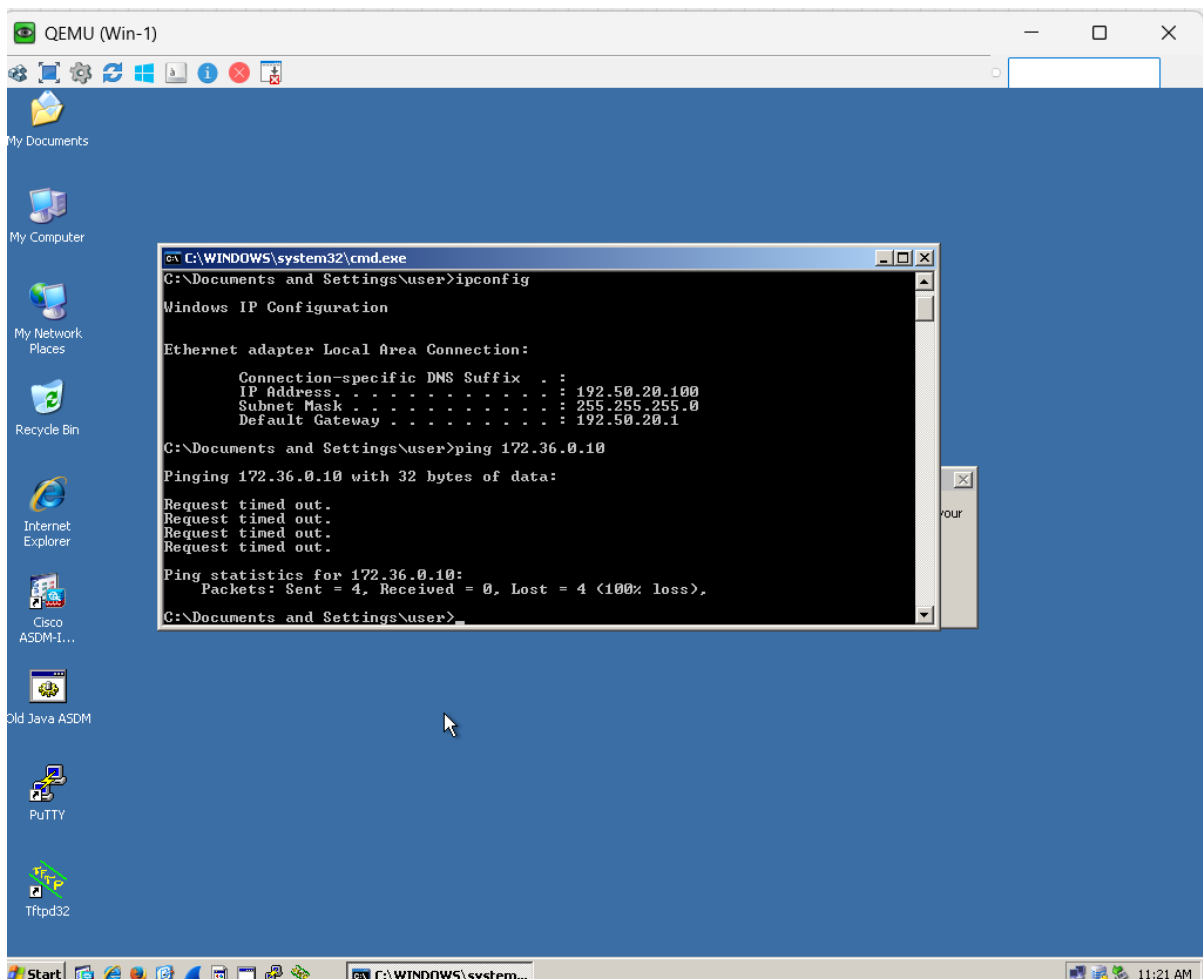
```
C:\WINDOWS\system32\cmd.exe
C:\Documents and Settings\user>ping 192.50.20.100
Pinging 192.50.20.100 with 32 bytes of data:
Reply from 192.50.20.100: bytes=32 time=3ms TTL=124
Reply from 192.50.20.100: bytes=32 time=2ms TTL=124
Reply from 192.50.20.100: bytes=32 time=2ms TTL=124
Reply from 192.50.20.100: bytes=32 time=2ms TTL=124
Ping statistics for 192.50.20.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms
C:\Documents and Settings\user>ping 223.0.50.10
Pinging 223.0.50.10 with 32 bytes of data:
Reply from 223.0.50.10: bytes=32 time=2ms TTL=124
Reply from 223.0.50.10: bytes=32 time=2ms TTL=124
Reply from 223.0.50.10: bytes=32 time=1ms TTL=124
Reply from 223.0.50.10: bytes=32 time=2ms TTL=124
Ping statistics for 223.0.50.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms
C:\Documents and Settings\user>
```

TASK 4: IDENTIFYING THE ERRORS

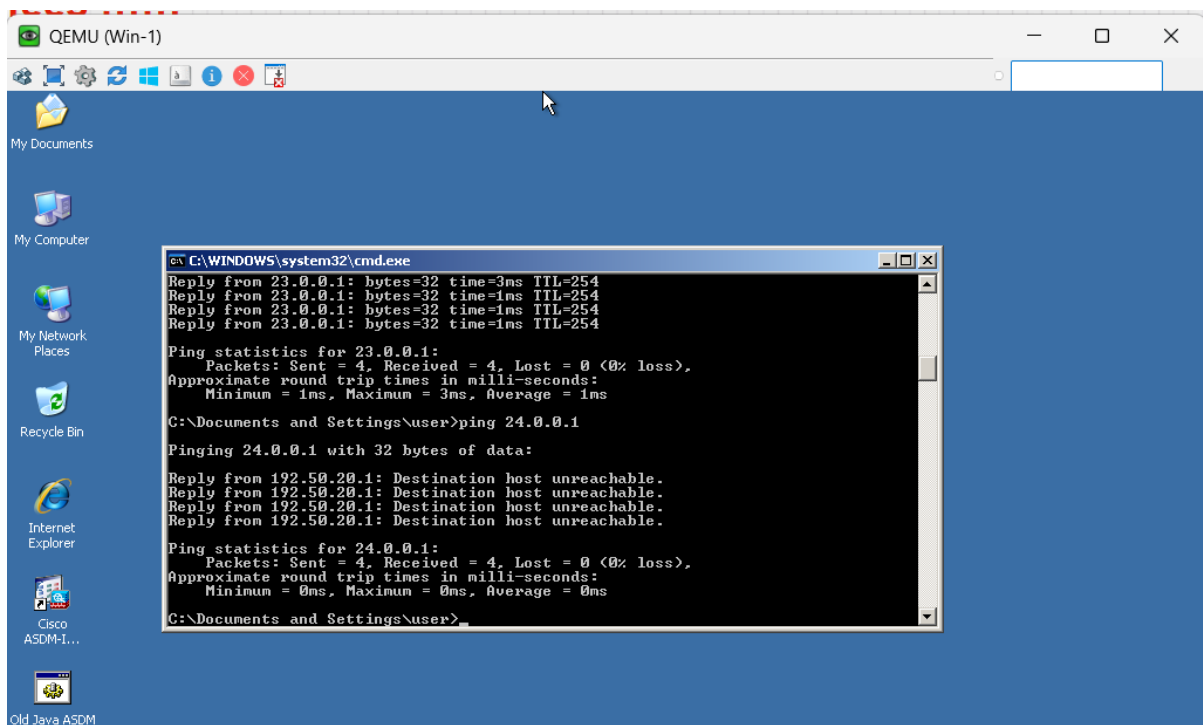
ERROR 1: Destination host unreachable. -> *There is no route for the Destination IP in the source PC route table.*



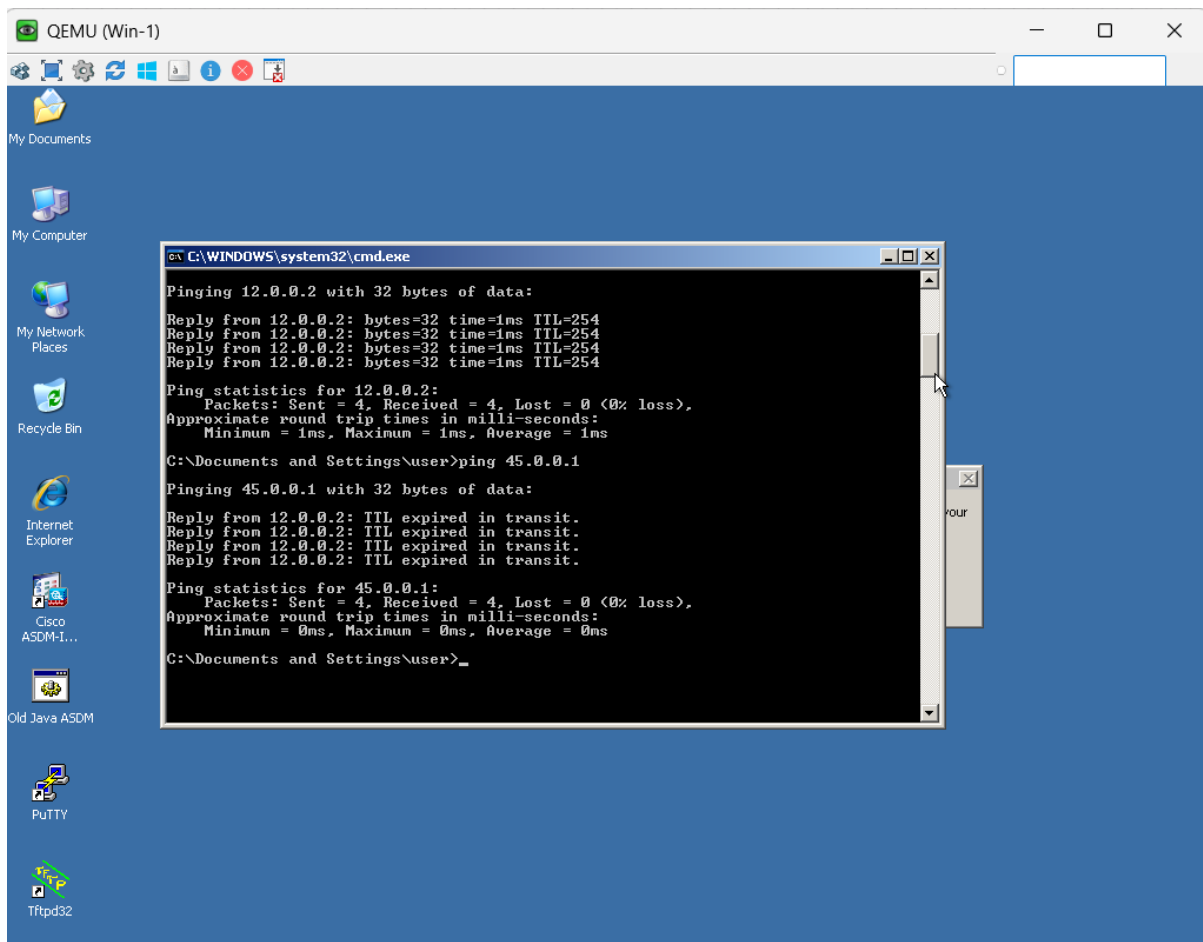
ERROR 2: Request timed out. -> *Data Packet sent, but no reply was received within the expected time.*



ERROR 3: Reply from 192.50.20.1, Destination host unreachable. -> *The router (192.50.20.1) responded that it could not reach the destination host.*



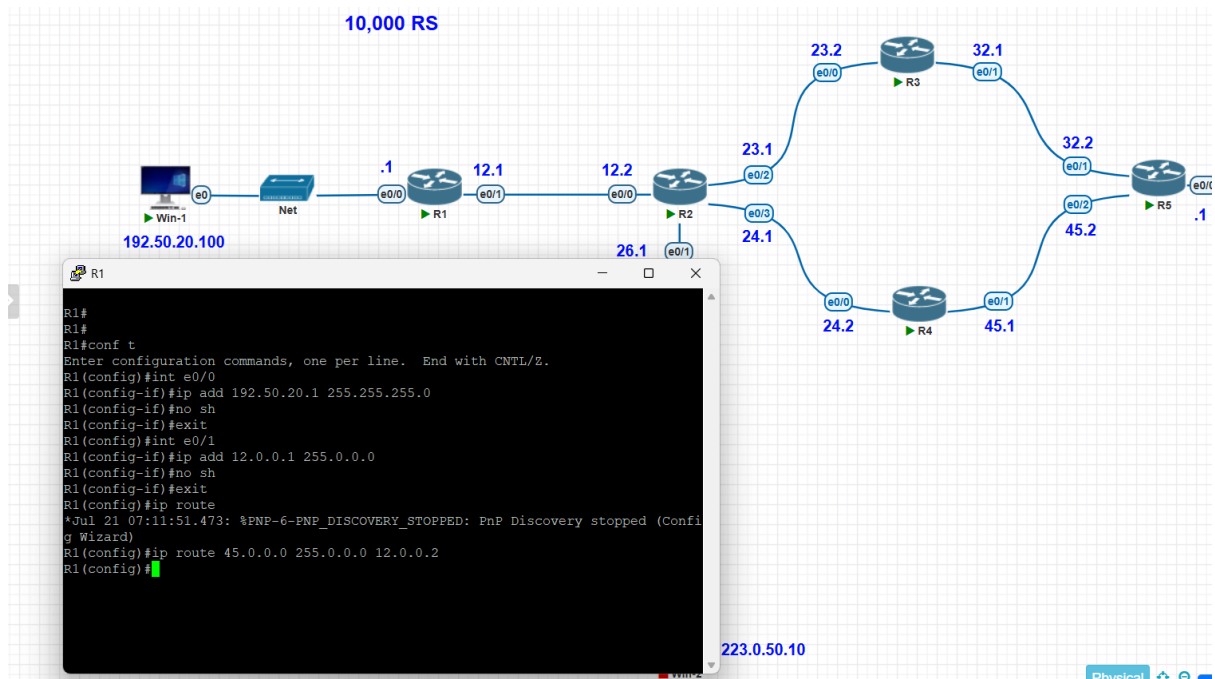
ERROR 4: Reply from 12.0.0.2, TTL expired in transit. -> *The packet exceeded its maximum hop limit (TTL) before reaching the destination and was discarded.*



HOW TTL OCCURRED?

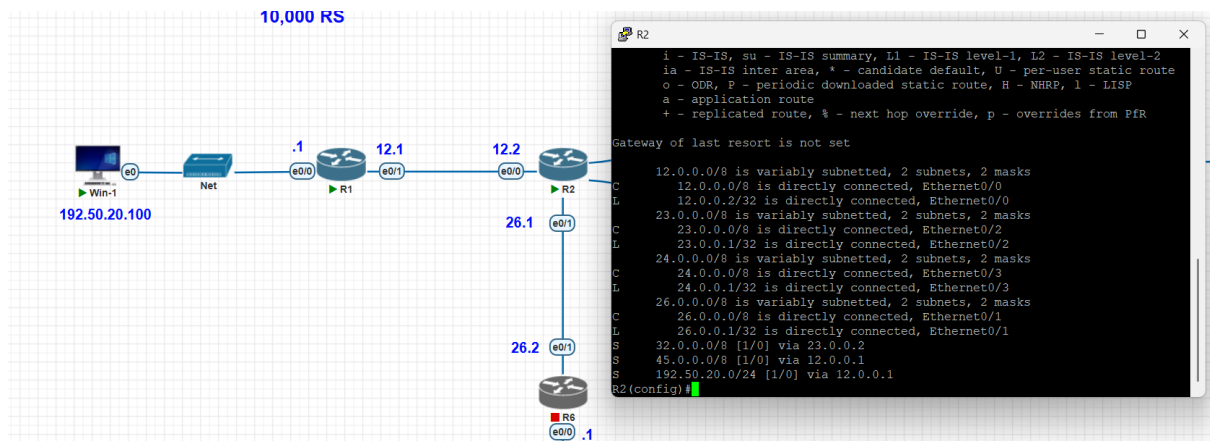
ROUTER 1 CONFIGURATION

In R1 the ip route for 45.0.0.0 is 12.0.0.2 (R2)



ROUTER 2 CONFIGURATION

In R2 the ip route for 45.0.0.0 is 12.0.0.1 (R1)



So, the TTL expired in transit error is occurred.